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E-commerce Group Assignment: Bookstore E-Commerce Website

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Introduction

The advancement of web technologies and the increasing reliance on online platforms have significantly transformed how users access products and services. E-commerce systems, in particular, have become essential for providing convenient, fast, and accessible shopping experiences. Traditional bookstores are often constrained by physical location, limited operating hours, and manual inventory management, which can reduce accessibility and efficiency. As a result, there is a growing demand for digital bookstore platforms that enable users to explore, search, and manage book purchases effectively.

Unity Bookstore is a web-based e-commerce application developed to address these challenges by offering a modern, interactive, and user-friendly digital bookstore experience. The system allows users to browse a curated collection of books, perform real-time searches, manage a shopping cart, and interact with a visually consistent and fully responsive interface across different devices. Strong emphasis was placed on usability, responsiveness, and client-side functionality to ensure a seamless user experience.

This project was developed as a group assignment to apply theoretical knowledge of web development in a practical and meaningful way. Built using Bootstrap 5, Vanilla JavaScript, and CSS3, the application demonstrates core front-end development concepts such as responsive design, dynamic content rendering, user interaction handling, theme customization, and data persistence using browser storage. Through this project, the team aimed to enhance practical skills in designing scalable, maintainable, and user-focused web applications while adhering to modern web development standards.

Project Objectives

General Objective

To design and implement a **user-friendly and responsive web-based bookstore** that provides an interactive platform for users to browse, search, and explore books efficiently. The system is designed to offer advanced filtering and categorization features, ensuring that users can easily find books of interest.

Specific Objectives

- Creating a **responsive homepage** using Bootstrap.
- Organizing books into clear **categories**: Education, Fiction, Research, and Children's Books.
- Designing **detailed book pages** with cover, synopsis, author, and price.
- Implementing a **JavaScript-based search bar** for real-time filtering by title and author.
- Providing a **functional contact form** for user feedback.

This project aims to combine modern front-end development techniques with practical usability features, resulting in a scalable, maintainable, and engaging digital bookstore platform..

Problem Statement

Despite the increasing demand for online shopping platforms, many traditional bookstores still rely on physical stores and manual processes for displaying, managing, and selling books. These limitations restrict accessibility, especially for users who cannot visit bookstores due to location, time constraints, or limited availability of information. Customers often face difficulties in searching for specific books, comparing prices, and receiving immediate feedback on availability.

In addition, the absence of interactive and responsive digital systems reduces user engagement and convenience. Many existing platforms lack efficient search functionality, proper catalog organization, and persistent shopping cart management. Therefore, there is a need for a modern, responsive, and user-friendly e-commerce bookstore system that enables users to browse, search, and manage book purchases efficiently while providing a consistent experience across different devices.

Project Selection & Justification

- Project Choice: **:** Bookstore E-Commerce Website
- Platform Selection: Bootstrap (HTML/CSS/JS)
- Justification: In this project we used custom coding over a CMS (like WordPress) to demonstrate core competencies in front-end development. Using Bootstrap allowed for a

mobile-first responsive design, while custom JavaScript enabled us to handle complex features like real-time filtering and a multi-theme engine without the overhead of heavy plugins.

The Tech Stack

- Structural Layer: HTML5 using Semantic tags for SEO.
- Presentational Layer: CSS3 with Custom Properties (Variables) and Bootstrap 5.3.
- Behavioral Layer: Vanilla JavaScript (ES6+), utilizing Classes for state management.
- Storage Layer: Browser local Storage API for session persistence.

Bootstrap was used to ensure a mobile-first responsive layout, while Vanilla JavaScript handled dynamic content rendering and application logic without reliance on external frameworks.

User Interface and User Experience Design

We implemented a Mobile-First Design approach. The layout uses a 12-column fluid grid system that re-stacks elements for mobile users while maintaining a professional grid for desktop users.

User Interface (UI/UX)

- **Consistency**: Used a unified color palette and typography (Inter and Playfair Display).
- **Feedback**: The "Add to Cart" function triggers a Bootstrap Modal to confirm the action to the user.
- **Responsiveness**: Utilized Bootstrap's grid system (col-lg-3 col-md-6) to ensure the catalog looks perfect on both desktop and mobile devices.

Website Structure

The website is organized into a logical flow to minimize clicks for the user:

1. **Homepage (`index.html`)**: Hero banner, search functionality, and featured books.
2. **Product Catalog (`books.html`)**: Full list of 20+ books with category and price filtering.
3. **Product Details (`product.html`):** Detailed view with descriptions and reviews.
4. **Shopping Cart (`cart.html`)**: Item management and checkout simulation.
5. **About Us (`about.html`)**: Project background and academic info.
6. **Contact Us (`contact.html`)**: User feedback form.

Functional Specifications

- **Real-Time Search & Clear:** The main-search bar on the home page allows users to query the book array by title or author. It includes a "Clear" functionality to reset the view.
- **Dynamic Catalog**: Books are stored as an object array in script.js. The DOM is updated dynamically, meeting the requirement of at least 20 unique items.
- **Theme Engine**: A custom Theme Manager class in theme.js allows users to switch between Light, Dark, Wood, and Forest themes using Local Storage for persistence.
- **Cart Simulation**: Uses local Storage to persist data across pages. It calculates subtotal, tax (10%), and shipping ($5.99 or free over $50).
- **Animations**: Integrated AOS (Animate On Scroll) for a premium, modern feel.

Database/Data Schema

Since we focused this on the front-end of our project, data is stored in a structured JSON-like format within script.js.

Field	Type	Description
id	Integer	Unique identifier for each book
title	String	The name of the book
category	String	Education, Fiction, Research, or Children
price	Float	Cost in USD
featured	Boolean	Determines if it appears on the home page

The project follows a modular structure to ensure maintainability:

- HTML5: Semantic structure for SEO and accessibility.
- CSS3 (Style.css): Custom properties (Variables) used for the theming system.
- JavaScript (script.js & theme.js):
  - script.js: Handles cart logic, book rendering, and search.
  - theme.js: Handles the premium theme switcher and particle effects.

Quality Assurance (QA) & Testing

Feature	Test Case	Expected Result	Result
Responsiveness	Resize to 375px (iPhone)	Grid collapses to 1 column	PASS
Search Logic	Type "Algo"	Shows "Introduction to Algorithms"	PASS
Reset Logic	Click "Clear" icon	Search bar empties and all books return	PASS
Cart Persistence	Add book & refresh page	Cart count remains at 1	PASS
Form Validation	Submit empty contact form	Browser shows “Please fill fields”	PASS

While the current version is a robust MVP (Minimum Viable Product), future iterations could include:

Future Enhancements

1. **Backend Integration**: Migrating the books array to a MongoDB or SQL database.
2. **User Authentication**: Implementing Firebase for user login and order history.
3. **Payment API:** Integrating Stripe or PayPal for real-time transactions.

- In general The Unity Bookstore project serves as a comprehensive example of a modern e-commerce front-end. By combining the layout power of Bootstrap with the logic of JavaScript, we have created a platform that is not only functional but also adaptable to user preferences through advanced theming and real-time data filtering.